Advisory

December 12, 2025

Subject:

Standardized underground installation advisory for services to single-family dwelling units

Background:

This standardized underground installation advisory has been developed to ensure safe, high-quality installations of underground electrical services within the City of Calgary. The purpose of this guide is to provide best-practice installation methods for electrical contractors and home builders.

Advisory:

Preferred Service Conduit Installation (Exhibit A):

- The service conduit shall be 63 mm rigid PVC.
- The conduit shall be installed from the bottom of the meterbase, vertically down to the foundation footing. The conduit shall be installed along the top of the foundation footing, in continuous contact, until reaching the building's side end, then continue in direct contact with the earth to the property line.
- Where abrupt elevation changes occur and the conduit is not in direct contact with the foundation or undisturbed earth, additional support is required. Install 2"x6" treated planking, securely strapped to the conduit, along its underside. (CE Code Rule 12-1106).
- Rigid two-hole straps shall be secured to the foundation, spaced no more than 1.8 m apart, along the entire vertical portion of raceway (CE Code Rule 12-114 1) d)).
- A rigid PVC expansion joint shall be installed between the meterbase and finished grade (CE Code Rule 12-012 13)).
- Minimum cover requirements for direct-buried service entrance raceways must meet the requirements as shown in Table 53 of the Canadian Electrical Code (CE Code Rule 12-012 1)).
- The exposed supply end of the conduit must be sealed to prevent moisture entry (CE Code Rule 6-300 3)).

Alternative Service Conduit Installation (Exhibit B):

- The service conduit shall be 63 mm rigid PVC.
- The conduit shall be installed from the bottom of the meter base, vertically down the foundation wall, to a rigid PVC 90° elbow. From the elbow, the conduit shall be installed horizontally along the foundation wall to the building's side end, then continue in direct contact with the earth until it reaches the property line.
- Where abrupt elevation changes occur and the conduit is not in direct contact with the foundation or undisturbed earth, additional support is required. Install 2"x6" treated planking, securely strapped to the conduit, along its underside. (CE Code Rule 12-1106).
- Rigid two-hole straps shall be secured to the foundation, spaced no more than 1.8 m apart. A rigid two-hole strap shall be installed directly below the bell end of the 90° elbow (CE Code Rule 12-114 1) d)).
- A rigid PVC expansion joint shall be installed between the meterbase and finished grade (CE Code Rule 12-012 13)).
- Minimum cover requirements for direct-buried service entrance raceways must meet the requirements as shown in Table 53 of the Canadian Electrical Code (CE Code Rule 12-012 1)).
- The exposed supply end of the conduit must be sealed to prevent moisture entry (CE Code Rule 6-300 3)).

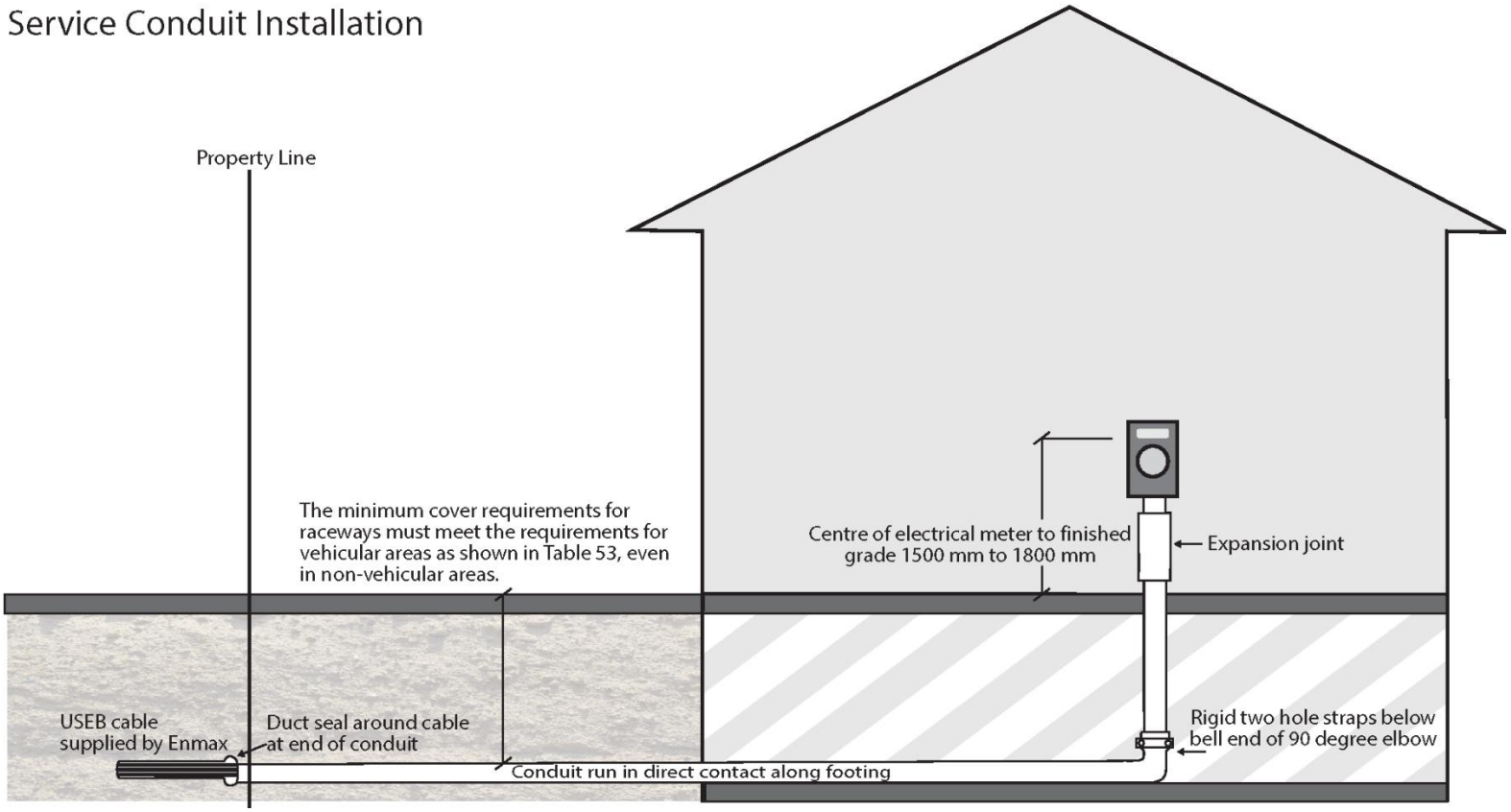
Grounding Requirements:

- The system grounding conductor shall be at least #6 AWG copper or #4 AWG aluminum, installed from the bonded neutral bar in the service section of the main distribution panel to the grounding electrode, ensuring a permanent and secure connection (CE Code Rule 10-114 1)).
- The ground plate shall be installed in direct contact with the exterior soil at a depth of at least 600 mm below grade (CE Code Rule 10-102 2) c)).
- The portion of the system grounding conductor located above grade shall be enclosed in rigid PVC conduit (CE Code Rule 10-116 3)).

For more information or questions, please contact the Electrical Technical Assistance Centre at electricaltac@calgary.ca

Exhibit A

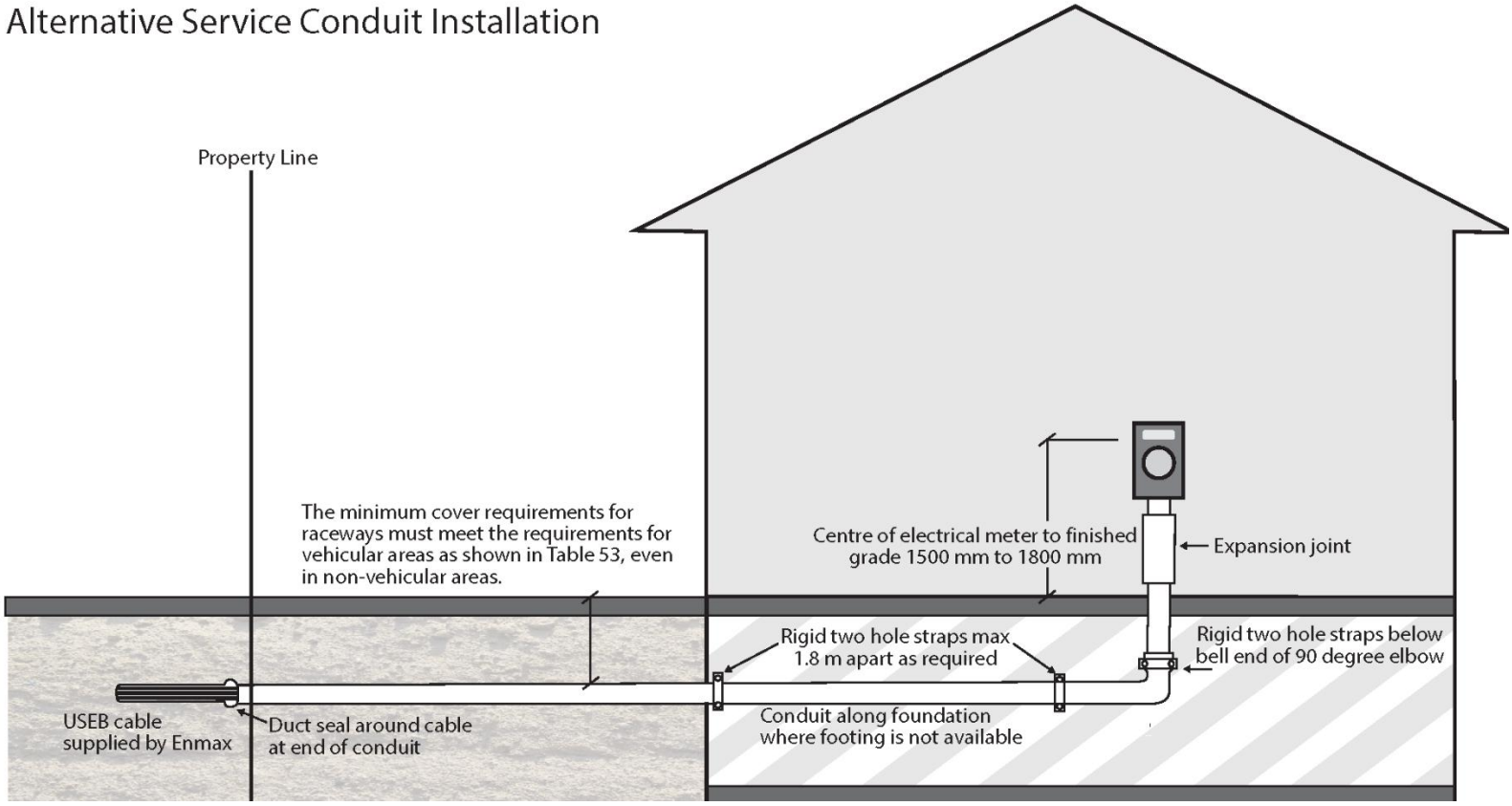
Service Conduit Installation



- **Note: USEB is an underground service entrance rated cable, supplied by ENMAX.**

Exhibit B

Alternative Service Conduit Installation



- **Note: USEB is an underground service entrance rated cable, supplied by ENMAX.**

This Advisory Bulletin is issued by The City of Calgary, Development, Business and Building Services. Advisory Bulletins have no legal status and cannot be used as an official interpretation of the various bylaws, codes and regulations currently in effect. The City of Calgary accepts no responsibility to persons relying solely on this information.